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Investigating the reliability of aggregate measurements of learning process data: From theory to practice

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Abstract

Background: Learning analytics (LA) research often aggregates learning process data to extract measurements indicating constructs of interest. However, the warranty that such aggregation will produce reliable measurements has not been explicitly examined. The reliability evidence of aggregate measurements has rarely been reported, leaving an implicit assumption that such measurements are free of errors.

Objectives: This study addresses these gaps by investigating the psychometric pros and cons of aggregate measurements.

Methods: This study proposes a framework for aggregating process data, which includes the conditions where aggregation is appropriate, and a guideline for selecting the proper reliability evidence and the computing procedure. We support and demonstrate the framework by analysing undergraduates' academic procrastination and programming proficiency in an introductory computer science course.

Results and Conclusion: Aggregation over a period is acceptable and may improve measurement reliability only if the construct of interest is stable during the period. Otherwise, aggregation may mask meaningful changes in behaviours and should be avoided. While selecting the type of reliability evidence, a critical question is whether process data can be regarded as repeated measurements. Another question is whether the lengths of processes are unequal and individual events are unreliable. If the answer to the second question is no, segmenting each process into a fixed number of bins assists in computing the reliability coefficient.

Major Takeaways: The proposed framework can be a general guideline for aggregating process data in LA research. Researchers should check and report the reliability evidence for aggregate measurements before the ensuing interpretation.

KEYWORDS

aggregation, internal consistency, learning analytics, learning process data, test–retest reliability

1 | INTRODUCTION

Learning analytics (LA) research has aggregated process data to extract measurements (or features) as indicators of constructs of interest to make sense and support students' engagement and self-regulated learning behaviours (Gray & Bergner, 2022; Munshi et al., 2023; Saqr & López-Pernas, 2021). For instance, LA dashboards may visualize the aggregate measurements to assist self-monitoring and facilitate reflection (Schwendimann et al., 2017). The measurements may serve as the input of machine learning models for detecting engagement and affect or predicting learning outcomes (Cui et al., 2019; Jiang et al., 2018). Clustering algorithms may use the measurements to group learners because the measurements characterize learning behaviours and strategies (Fincham et al., 2018; Jovanović et al., 2017).

Aggregating process data is a common practice in LA because raw process data may be difficult to understand due to their complexity and size. Despite its wholesale application, the psychometric rationale of such aggregation has not been explicitly considered. Winne (2020) states that the granularity of aggregate measurements should be neither too fine to cause unreliable individual differences nor too coarse to mask within-subject variances. Building on this statement, this paper examines the pros and cons of aggregation in terms of psychometrics and proposes caveats for aggregating process data.

Additionally, the validity and reliability evidence of aggregate measurements is rarely reported in LA research. Researchers have started investigating the validity issue of measurements in LA and recommended methods to improve the validity, such as grounding the measurements in theories, data triangulation, and combining data-driven and theory-driven approaches (Fan, van der Graaf, et al., 2022; Motz et al., 2019; Winne, 2020). In contrast, the reliability issue is more underexplored, but the reliability of measurements is as important as its validity. Measurements with poor reliability may decrease the statistical power and increase the false discovery rate in group difference analyses and correlation analyses (Parsons et al., 2019). Ignoring the reliability evidence of aggregate measurements implies that such measurements are free of errors. We argue that this assumption does not hold. Even physical measurements contain errors (Bergner, 2017). Accordingly, this paper investigates ways to examine the reliability of measurements derived from aggregating process data.

2 | LITERATURE REVIEW

2.1 | The pros and cons of aggregation

Aggregation, known as item parcelling in latent variable modelling, refers to using the sum or mean of two or more items as an observed measurement of the construct of interest (Little et al., 2002). Aggregation is widely used in social science because of its advantages in psychometric properties and model estimation (Little et al., 2002; Little et al., 2013; Matsunaga, 2008). Compared with item-level measurements, aggregate-level measurements are more reliable and follow distributions closer to normality and continuity. Latent variable models with aggregate-level indicators are more parsimonious and stable and have fewer spurious residual correlations and factor loadings. However, improper aggregation may lead to incorrect interpretation of construct dimensionality and mask model misspecification. Thus, aggregation should be based on careful consideration. Little et al. (2002) suggested two key issues for such consideration (dimensionality and research purpose), which we discuss in the context of aggregating process data in section 3.2.

2.2 | Measurements based on aggregating learning process data

Process data represented as event sequences are common in LA, such as action logs, think-aloud data, and eye movement data. Such data contain rich information about the temporality of learning. Researchers have suggested three types of measurements of temporality (Hadwin, 2021; Knight et al., 2017; Molenaar, 2014): (1) individual, (2) co-temporal, and (3) sequential characteristics. Individual characteristics are the time-related characteristics of individual learning events (Knight et al., 2017; Molenaar, 2014), such as the frequency (how many times a particular event happens) and timing (when a particular event occurs). Co-temporal characteristics refer to multiple events co-occurring within a window, regardless of in which order (Hadwin, 2021). In contrast, sequential characteristics, also named temporal order, are concerned with the order of multiple events in time (Chen et al., 2018; Fan, Tan, et al., 2022). Examples are patterns of ordered actions that frequently occur during an educational game.

Regardless of the temporal characteristic of learning, aggregation is often an essential step to derive meaningful measurements to summarize the learning process. For individual characteristics, aggregation produces the frequencies of individual events. For co-occurrence and sequential characteristics, aggregation counts the frequencies of patterns of events and uses the frequency to derive other measurements, such as transition metrics, which quantify the tendency of one event following another (Bosch & Paquette, 2021). The granularity of aggregate measurements varies across research. Some studies have aggregated the whole learning process, even though the process lasted months (Cicchinelli et al., 2018; Li et al., 2020). Other studies have segmented the process into sessions and aggregated each session (Fincham et al., 2018; Jovanović et al., 2017). Nevertheless, few studies have explicitly considered the rationales of such aggregation (Winne, 2020). Moreover, the psychometric property of aggregate measurement is hardly examined, leaving an implicit assumption that the measurement is free of errors.

2.3 | Reliability in standard assessment

In standard assessment, reliability refers to the consistency of assessment scores across multiple measures of the same construct (AERA et al., 2014). Generally, there are two ways to express reliability – the reliability coefficient and the standard error of measurement (SEM). The reliability coefficient quantifies the consistency of assessment scores by using an index of the amount of true variance operating in a set of raw scores. SEM is an index of score differences due to measurement errors. For test takers, it shows the average difference between the observed scores and their "true" scores (scores without any measurement error). This paper focuses on applying the reliability coefficient to aggregate measurements of process data because it is more widely used and familiar to LA researchers and more accessible to compute than SEM.

One type of reliability coefficient is internal consistency, which quantifies the extent to which assessment items measure the same construct (Webb et al., 2006). Coefficient α, sometimes named Cronbach's α, is often misinterpreted as a measure of internal consistency. While it is related to internal consistency, it is heavily influenced by the number of items and is not a robust measure of internal consistency (Cortina, 1993; Yang & Green, 2011). A measurement with low internal consistency may show a high α due to many items. This is particularly relevant in the LA context because there are often many events in a learning process, analogous to many items. Alternatives, such as coefficient ω, have been recommended as a more robust measure of internal consistency (Flora, 2020; Hayes & Coutts, 2020).

Split-half reliability is another type of internal consistency measure (Thorndike & Thorndike-Christ, 2010). It is obtained by dividing the assessment into equivalent halves (e.g., first and second halves as well as odd and even halves) and correlating the scores from both parts. A highly reliable assessment requires a high correlation, indicating the participant's scores are consistent on both halves of the assessment.

Another type of reliability coefficient is test–retest reliability, on the condition that the identical assessment is administrated on multiple occasions, typically with a short interval between occasions (Thorndike & Thorndike-Christ, 2010). According to Cronbach (1972) and Thorndike (1947), the lasting and general aspects of test takers, such as their ability, will contribute to the consistency of assessment scores when the assessment is administered on different occasions. By contrast, temporal and context-specific factors (e.g., noisy environment and possible disruptions from other test takers) cause transient errors in assessment scores and undermine consistency (Becker, 2000). Test–retest reliability coefficients account for transient errors, but internal consistency does not (Leppink & van Merriënboer, 2015; Schmidt et al., 2003).

Common test–retest reliability coefficients are linear correlations and intraclass correlations (ICC) (McGraw & Wong, 1996). Linear correlations are limited to cases where an assessment is conducted on exactly two occasions, while ICC applies to multiple occasions. There are multiple versions of ICC, and readers may refer to McGraw and Wong (1996) as well as Koo and Li (2016) for the details of various versions.

3 | A FRAMEWORK OF AGGREGATING PROCESS DATA

3.1 | Events as item responses

In standard psychological and educational assessments (e.g., aptitude tests and personality and attitude scales), the basic measurement unit is items/questions, and the construct of interest is inferred based on subjects' responses to items (Mislevy et al., 2012). For instance, students' responses to addition problems indicate their numeracy proficiency. When we apply process data to measure constructs of interest, the basic unit may be events (Reimann, 2009). We infer the construct of interest based on the individual, co-occurrence, and sequential characteristics of events. For instance, students' academic procrastination has been operationalized as the interval between when they start an assignment and when the assignment is released (Huang et al., 2021). This operationalization involves the timing characteristic of the event of starting an assignment. Similarly, behavioural engagement in digital learning environments has been inferred based on the frequency of clicks and posts (Saqr & López-Pernas, 2021). In this sense, a learning event may be analogous to an item response in standard assessment.

The response to a single item is unreliable because of item uniqueness and measurement errors (Little et al., 2002; Rushton et al., 1983). So is the characteristic of a single event (Winne, 2020). Moreover, the measurement errors of events may be much larger because events in LA usually naturally occur and are not intentionally triggered to measure a construct. Taking procrastination as an example, we may not infer a student's procrastination tendency based on the time that they started a single assignment. Additional factors, such as a student's estimate of time for the assignment, may influence the decision of when to start this assignment and the interval between the starting time and the assignment release time. However, within a week, the procrastination tendency may be relatively stable across assignments, while the additional factors may fluctuate. Aggregating the intervals across assignments may mitigate the influence of the additional factors, and the aggregate interval may be more indicative of procrastination. Thus, researchers usually aggregate events over a period to reduce measurement errors and improve reliability (e.g., Saqr & López-Pernas, 2021). Nevertheless, the warranty of such aggregation has not been explicitly examined, which we discuss in the next section.

3.2 | Aggregation on process data

Little et al. (2002) suggested two key issues for the aggregation decision: dimensionality and research purposes. Aggregation is warranted when items involve the same construct or the same facet of a construct, but it is risky when items involve different constructs or different facets of a construct. Aggregation is beneficial when the research purpose is to understand the latent construct and its relations with other constructs, but it should be avoided if the purpose is to understand the relations among items. Both issues may be mild in the context of aggregating process data. First, the aggregation is usually conducted on events of the same category. That is, an aggregate measurement is based on events involving the same construct or the same facet of a construct. It is worth noting that this property relies on the prerequisite that events are a valid indicator of the construct. For example, when studying self-regulation, if events indicative of evaluation are incorrectly classified as reflection, the aggregate measurement of reflection would be a mixture of reflection and evaluation events. Consequently, this measurement would be invalid and its reliability would be meaningless.

For the second issue, LA researchers are interested in knowing the relations between constructs and utilizing such understanding to improve the learning process (Wise et al., 2021) rather than the relations among individual events indicating these constructs. For instance, it may be helpful to know whether students who usually start assignments early will obtain higher average assignment scores than those starting assignments near the deadline, but the relation between the time of starting the first assignment and the score on this assignment is trivial.

One issue not covered by Little et al.'s (2002) suggestion but critical in process data is the temporal nature of the learning process. Process data are generated over a period. The reliability of aggregate measurements may increase as the period becomes longer (in a more general term, the analytic unit or granularity becomes coarser), given that longer periods contain more events indicative of the same construct of interest, and the construct is stable within the period. However, when the period is relatively long, changes may occur in learning behaviours, which represents temporal within-subject variation (Saqr, 2023). Such variation can be useful for understanding behavioural adaptation (Williams et al., 2021). In terms of self-regulated learning, students can exercise human agency in their learning and adapt behaviours based on learning goals and current progress (Martin, 2004). The temporal variation in learning events may capture behavioural adaptation (Greene et al., 2021). However, aggregation over a relatively long period will mask this temporal variation (Winne, 2020), failing to reveal behavioural adaptation.

Distinguishing temporal variation and transient errors mentioned in the discussion about test–retest reliability avoids confusion. Transient errors, in nature, are a special case of temporal variations. However, it is reasonable to assume that learning behaviours are stable over a relatively short period. Within this period, the temporal variation in behaviours is transient errors (Schmidt et al., 2003). In contrast, assuming stable behaviours over a relatively long period may be unreasonable. In such cases, temporal variation should be studied as it may be significant and meaningful (Schmidt et al., 2003). For example, students starting the first assignment early may start the second assignment early but be less likely to start the final assignment, taking place months later, early as well.

Deciding the period for aggregation belongs to a more general issue, that is, the granularity of the analytic unit. The proper granularity and period for aggregation depend on research contexts and questions. In a course spreading over months, students' learning behaviours may not significantly change within a few weeks (Jovanović et al., 2017). By contrast, in an intense learning activity, meaningful changes may occur in learning behaviours within hours. For example, in a programming task of one hour, Berland et al. (2013) found that novices' programming activities generally went through three phases: exploration, tinkering, and refinement.

Even when researchers are only interested in the overall relationship between two constructs rather than their temporal trajectories, longitudinal data analysis may produce more reliable results. The literature on longitudinal data analysis has well discussed this issue (e.g., see chapter five in Weiss, 2005). As an illustration, Leppink and van Merriënboer (2015) showed that aggregating the repeated measurements of cognitive load might result in a failure to reveal the positive effect of a supportive tool in lowering the cognitive load.

3.3 | Reporting evidence of reliability for aggregate measurements

Regardless of the period length (more generally, the granularity), providing evidence of reliability for aggregate measurements would be beneficial. Researchers have considered reliability for some aspects of process data analysis. For example, when raw events are manually coded into categories based on a theoretical framework, the inter-rater agreement is provided as evidence of coding reliability (e.g., Chen et al., 2017). Similar attention to reliability should be extended to aggregate measurements of process data. Here we outline approaches to estimating and reporting the reliability evidence of aggregate measurements. Figure 1 displays the core recommendation in a flow chart.

3.3.1 | Are the process data repeated measurements?

The first point to consider is whether the process data are repeated measurements, that is, whether events indicative of the same construct can be viewed as the responses to the same item at different time points or the responses to different items. In standard assessments, repeated measurements mean that identical external stimuli are presented to individuals. In LA, the learning condition is analogous to the stimulus. What consists of the learning condition depends on the construct of interest and the research context. For instance, in a MOOC course, if the construct is behaviour engagement operationalized as the action frequency, the learning condition may be conceptualized as the course policy and learning topic. A policy awarding for regular and frequent course activities incentivizes learners' actions, and topics appealing to a learner motivates their actions. The policy is generally stable throughout the course, but the topic may change each week. Consequently, the action frequencies in different weeks are more of responses to different items rather than the same item in different weeks. If the construct of interest is academic procrastination operationalized as the interval between the assignment release and submission, the learning condition may be the course policy, the time of a day for releasing an assignment, and the allowed time window for its completion. A policy awarding for early assignment completion incentivizes early submission. Given the time window allowed for completing an assignment, releasing it in the middle of the night results in an interval between its release and deadline different from releasing it at noon. If these conditions are consistent throughout a course, it may be reasonable to treat the intervals between the release and submission of different assignments as repeated responses to the same item. Note that the treatment assumes that the difficulty and complexity of assignments do not influence procrastination.

3.3.2 | Do different processes contain an equal number of reliable events?

Regardless of whether process data are repeated measurements, there are a few challenges for applying reliability coefficients to process data. The most apparent one is that the lengths of processes often vary across learners. For instance, the number of events in a MOOC course varies across students in a fixed period. This means

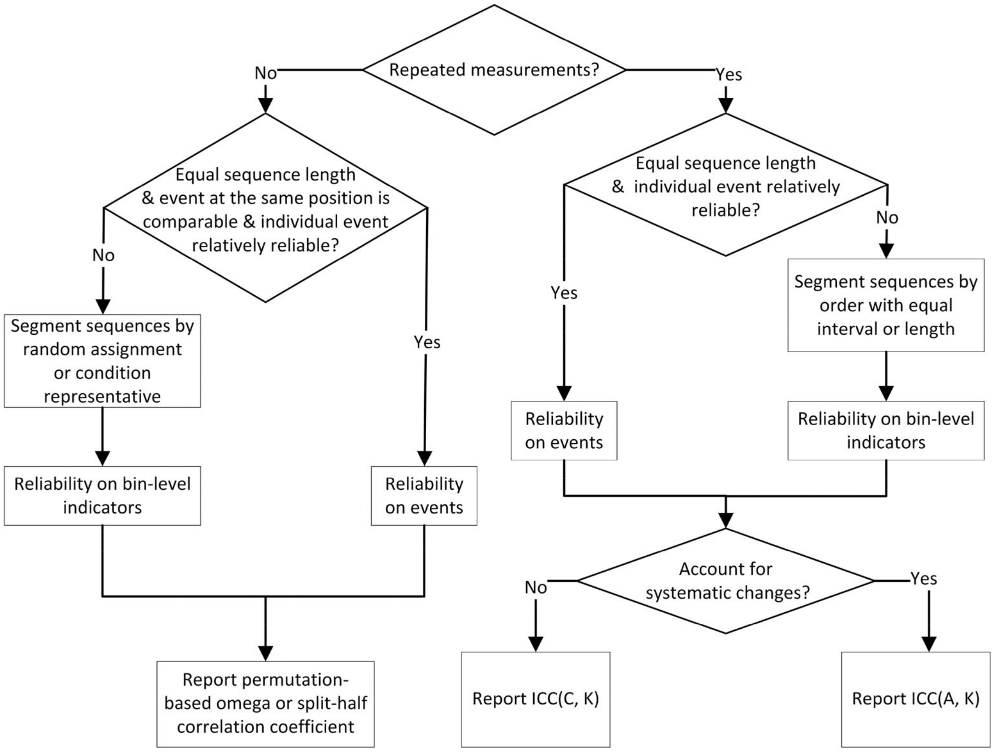

FIGURE 1 Decision flow for reporting reliability evidence of aggregate measurements.

that the number of responses varies across students. Another challenge is that the measurement errors of individual events can be large because the events in LA occur in a natural environment (Winne, 2020), where many random factors may excite or inhibit their occurrence. Computing reliability coefficients directly based on raw events may underestimate the true reliability. If process data are not repeated measurements, one more challenge is that events at the same position in different processes may be responses to different items. For instance, the second events of two students' learning processes in a MOOC course may occur in weeks one and two, respectively. Unless the learning condition is consistent in the two weeks, the students' second events are not comparable.

To overcome the above challenges, we need to convert different learners' process data to a fixed number of responses. One option is segmenting the process that will be aggregated into bins and regarding the measurement aggregated within a bin as a response. Bin-level measurements are more reliable than individual events because the influence of random factors may vary across events and be offset mutually in the bin-level measurements. The segmenting should be carefully carried out. Otherwise, variances unrelated to the construct of interest may cause the measurements to be inconsistent across bins and result in a poor reliability coefficient estimate. We propose four segmenting approaches below, which are illustrated in Figure 2.

By order

The most straightforward method may be segmenting a process into bins with equal lengths, that is, the same number of events (e.g., Kinnebrew et al., 2014). Alternatively, if different processes cover the same clock-based units (e.g., hours and days), segmenting can be done to create bins with equal intervals (e.g., one hour or one day as a bin).

Random allocation

This method refers to randomly allocating an event into a bin until all events are allocated. Different bins have equal lengths. To mitigate the impact of randomness in allocating events, we need to repeat the segmenting multiple times, compute the reliability estimate each time, and report the average estimate.

Condition-representative sampling

This approach is a special form of random allocation. It is analogous to stratified sampling and randomly allocates events occurring in the same condition to a bin so that each bin is representative of all conditions. This approach aims to ensure that the differences between bin-level measurements are mainly because of random factors and not heavily influenced by condition-specific effects (analogous to item-specific effects in test theory; McDonald, 1999). If different bins involve different conditions, the proportion of bin-level measurement variances shared across bins would decrease (Little et al., 2002). This issue may cause lower correlations between bin-level measurements and poorer reliability estimates. Random allocation also implicitly controls the condition-specific effect because the events in the same condition may be randomly assigned to different bins.

For process data that are not repeat measurements, random allocation and condition-representative sampling are recommended as they control the potential condition-specific effect. For those that are repeat measurements, by order with equal lengths or intervals is recommended because the segmenting preserves the transient errors

Condition	C1			C2						C3		
Day	Tuesday			Wednesday						Friday		
Event ID	c1n1	c1n2	c1n3	c2n1	c2n2	c2n3	c2n4	c2n5	c2n6	c3n1	c3n2	c3n3

Event sequence

The events in each bin

Segment approach	Bin 1	Bin 2	Bin 3
Equal interval	c1n1 : c1n3	c2n1 : c2n6	c3n1 : c3n3
Equal length	c1n1, c1n2, c1n3, c2n1	c2n2, c2n3, c2n4, c2n5	c2n6, c3n1, c3n2, c3n3
Random allocation	c1n2, c2n4, c3n2, c3n3	c1n1, c1n3, c2n3, c2n5	c2n1, c2n2, c2n6, c3n1
Condition-representative	c1n3, c2n1, c2n6, c3n2	c1n1, c2n2, c2n4, c3n3	c1n2, c2n3, c2n5, c3n1

FIGURE 2 An illustration of segmenting methods.

for each bin. Transient errors are "caused by momentary and nonrepeating factors" (Becker, 2000, p. 370). Preserving transient errors allows researchers to evaluate how stable the measurements are over time.

3.3.3 | Choosing the reliability coefficient

For process data that are not repeat measurements, the internal consistency coefficient should be provided as evidence of its reliability. We recommend coefficient ω because it has been widely recognized as a more robust measure of internal consistency than coefficient α (Flora, 2020; Hayes & Coutts, 2020), and tools for computing ω are available in common data analysis platforms, such as OMEGA macro for SPSS and SAS (Hayes & Coutts, 2020), the *psych* library in R, as well as the *reliabiliPy* library in Python. Flora (2020) provides a tutorial on selecting the proper version of ω. Alternatively, if the process data are too short and can only be segmented into two bins, the split-half correlation coefficient may be reported. However, obtaining an accurate estimate of this coefficient may necessitate an enormous amount of random segmenting (Parsons et al., 2019).

For process data that are repeat measurements, test–retest reliability coefficients should be reported since they account for transient errors (Schmidt et al., 2003). If ICC is chosen, it is worth noting that there are two distinct types of ICC: absolute agreement versus consistency. ICC for consistency focuses on whether the rankings of individuals' scores are consistent across occasions and does not regard the systematic score differences between occasions (i.e., mean differences) as transient errors. ICC for absolute agreement assumes perfect consistency and regards the systematic differences as transient errors. Consequently, within a dataset with high ICC for consistency, ICC for absolute agreement may be low. Koo and Li (2016) claim that ICC for an absolute agreement should always be chosen for test–retest reliability. We concur with Parsons et al.'s (2019) argument that it is up to researchers to select the version of ICC based on the importance of the systematic differences within the specific research context. This is relevant to process data because the change in learners' internal states may cause a systematic change in learning behaviours. For instance, it is well known that MOOC learners' engagement decreases as the course progresses (Bote-Lorenzo & Gómez-Sánchez, 2017). However, we may be interested in whether their engagement rankings are consistent between adjacent weeks or months. Beyond the above disagreement on ICC for absolute agreement versus consistency, there is a consensus that ICC based on the two-way mixed-effect model should be used for test–retest reliability (Koo & Li, 2016; Parsons et al., 2019). Following McGraw and Wong's (1996) notions, we used ICC(A, k) and ICC(C, k) to denote ICC based on the two-way mixed-effect model for absolute agreement and consistency, respectively.

3.4 | Summary

This section presents a framework for aggregating process data, which argues that events in learning processes are analogous to item responses but more unreliable. Aggregating events within a proper period are recommended to obtain more reliable measurements of constructs of interest. We recommend reporting reliability evidence for aggregate measurements, regardless of the period length. A general guideline is proposed for selecting the appropriate reliability coefficient and computing the estimate for aggregate measurements. Note that the guideline is not ultimate. Researchers should choose the reliability coefficient based on the research purpose and context. The next section illustrates the framework by applying it to compute the reliability estimates of aggregate measurements in the context of an introductory computer science course.

4 | CASE STUDY: AGGREGATE MEASUREMENTS OF ACADEMIC PROCRASTINATION AND PROGRAMMING PROFICIENCY

This section extracts aggregate measurements of academic procrastination and programming proficiency from submission sequences of computer programming assignments. We applied the proposed framework to compute reliability estimates to illustrate: (1) aggregate measurements are more reliable than individual events; (2) two events or measurements with a long period between them are weakly related and not suitable for aggregation; (3) how to segment event sequences; (4) how different segmenting approaches cause distinct estimates of reliability.

4.1 | Data and participants

Data were collected from an undergraduate introductory computer science course at a public university in the U.S. during the Fall 2019 semester. Six hundred and seventeen students completed the course and approved using their assignment data for research purposes. Among them, 30% were female, 69% were male, and 1% preferred not to report their gender. 81% were first-year students, and 45% were in a CS-related major.

The course taught Java programming and lasted 15 weeks. A web-based, problem-driven learning system hosted all homework assignments, lab exercises, and exams. The learning system released a minor programming assignment each weekday. Students had 24 h to submit solutions to an assignment after its release. The system automatically graded the submission and provided feedback if the submission had errors (e.g., syntactic and semantic errors). Students could submit unlimited times to reach a correct solution and obtain full credits within 24 h. After this period, students could still submit solutions but would not receive credit. There were 68 assignments in total. The learning system automatically recorded the code, date, correctness, and error feedback of all student submissions. We used the submission sequences to extract measurements of academic procrastination and programming proficiency and examined the measurement reliability.

4.2 | The reliability of aggregate measurements of academic procrastination

Academic procrastination is delaying an academic task toward the deadline or a failure to complete it before the deadline (Schraw et al., 2007). Two operationalizations have been used in technology-enhanced learning environments: (1) whether an assignment is completed before the deadline (Huang et al., 2021); (2) the interval between the time of releasing an assignment and the time students starting the assignment (Zhang et al., 2022). This study only adopted the second operationalization, given that the purpose is to illustrate the reliability of aggregate measurements. If a student did not start an assignment before the deadline, this student's procrastination in this assignment was set to be 24 h.

The course policy did not change over time, and all assignments were released at midnight. Thus, we regarded the learning condition as consistent throughout the course and viewed the time students started different assignments as repeated responses to the same procrastination item. Each student showed 68 responses. We firstly computed ICC(A, k) and ICC(C, k) of procrastination measurements aggregated over 2–34 responses to illustrate how the reliability changed as the period being aggregated lengthened and the difference between ICC(A, k) and ICC(C, k). We then compared the test–retest reliability of individual procrastination responses and week-level procrastination measurements.

4.2.1 | Results

Figure 3 shows that ICC(A, k) was slightly lower than ICC(C, k). This is expected as ICC(C, k) did not treat systematic differences between responses as transient errors. Both types of ICC increased as the number of procrastination responses being aggregated increased. Notably, from two to four responses, the average ICC(A, k) rose from 0.59 to 0.74, and the average ICC(C, k) rose from 0.62 to 0.76. After that, the growth rate of both ICC slowed down, indicating a diminishing benefit of aggregation. When the number of responses reached five, which was the number of assignments in most weeks, the average ICC(A, k) and ICC(C, k) were 0.76 and 0.78, indicating moderate to good reliability. Figure 4 clearly shows that the ICC of week-level indicators was higher than individual responses. Overall, these results suggest that aggregation over a short period improved the reliability of procrastination measurements.

Figure 5 depicts the correlations among individual responses (upper right) and week-level aggregate measurements of procrastination (lower left).[1] The farther away from the main diagonal a cell is, the lighter its shade is, indicating the diminishing associations among individual responses or aggregate measurements over time. For example, the average correlation between the individual responses within week 1 (A1–A5) was 0.31. By contrast, the average correlation was 0.22 between the individual responses of weeks 1 and 7 (A33–A37) and 0.14 between that of weeks 1 and 14 (A64–A68). The average correlation between adjacent week-level measurements was 0.73, while the average correlation between measurements with an interval of 7 weeks (e.g., week 1 and week 8, week 2 and week 9) was 0.51.

[1] Procrastination on A41 and A42 had a high correlation because these assignments were released at the same day.

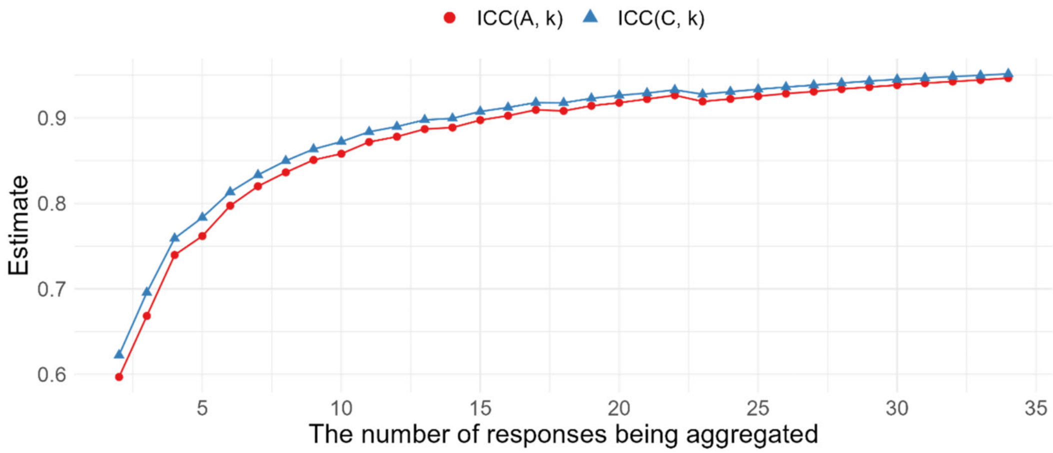

FIGURE 3 The average reliability estimates at each number of responses.

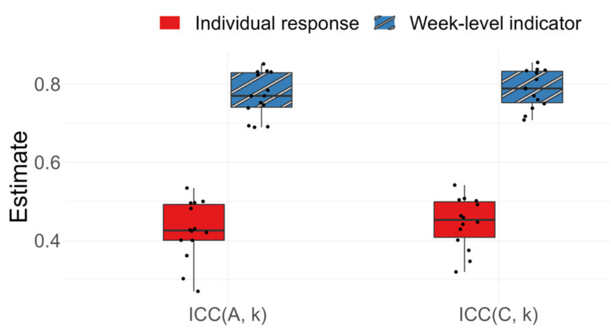

FIGURE 4 ICC of individual responses and week-level procrastination indicators.

Overall, the results suggest the instability of procrastination behaviour over long periods and the importance of selecting the appropriate period for aggregation.

4.3 | The reliability of aggregate measurements of programming proficiency

Programming proficiency is not a unidimensional construct, as completing a program entails various skills (Xie et al., 2019). The current study used four dimensions of programming proficiency to illustrate segmenting event sequences to compute the reliability of aggregate measurements: programming syntactic, programming semantic, syntactic error debugging, and semantic error debugging proficiency. Research has used the proportion of submissions to programming assignments containing syntax errors as a measurement of syntax proficiency (Zhang et al., 2023). We adopted this operationalization but acknowledged that this measurement might be just a facet of syntactic proficiency. Similarly, we operationalized (1) semantic proficiency as the proportion of submissions containing semantic errors, (2) syntactic debugging proficiency as the proportion of submission pairs where the second submission fixed at least one syntax error in the first submission, (3) semantic error debugging proficiency as the proportion of submission pairs where the second submission fixed at least one semantic error in the first submission.

Because students' programming proficiency was likely to develop throughout the course, we only used submission sequences in the last three weeks. We assumed that the growth rate of programming proficiency was relatively slow at the end of the course and that programming proficiency might be stable within a week. Thus, we computed the four programming proficiency indicators within each of the three weeks. As a validation, we correlated the week-level measurement with the final exam scores. The correlations were weak to moderate (r ranges from -0.52 to 0.30, with the average absolute value of 0.21, and all $p < 0.01$ after Bonferroni correction except for two correlations).

Even in the same week, assignments might differ in syntax and semantic complexity and involve different programming constructs. Thus, submissions were more of responses on different items rather than repeated responses on the same item. The number of submissions per student per week ranged from 10 to 420. Based on Figure 1, we segmented each submission sequence using both random allocation and condition-representative approaches. In the current context, the condition was programming assignments. We also segmented submission sequences by order with equal intervals and lengths to illustrate the condition-specific effect on the reliability estimate. All segmenting approaches split a sequence into three bins, and coefficient ω was computed. Since there was no assignment on either of Monday and Tuesday as well as Thursday and Friday in the last three weeks, the equal interval method allocated Monday and Tuesday to the 1st bin, Wednesday to the 2nd bin, as well as Thursday and Friday to the 3rd bin.

4.3.1 | Results

Figure 6 presents the ω estimates of aggregate measurements based on different segmenting methods. Because random allocation and condition-representative methods involve a random process (randomly assigning submissions to bins), we repeated each method

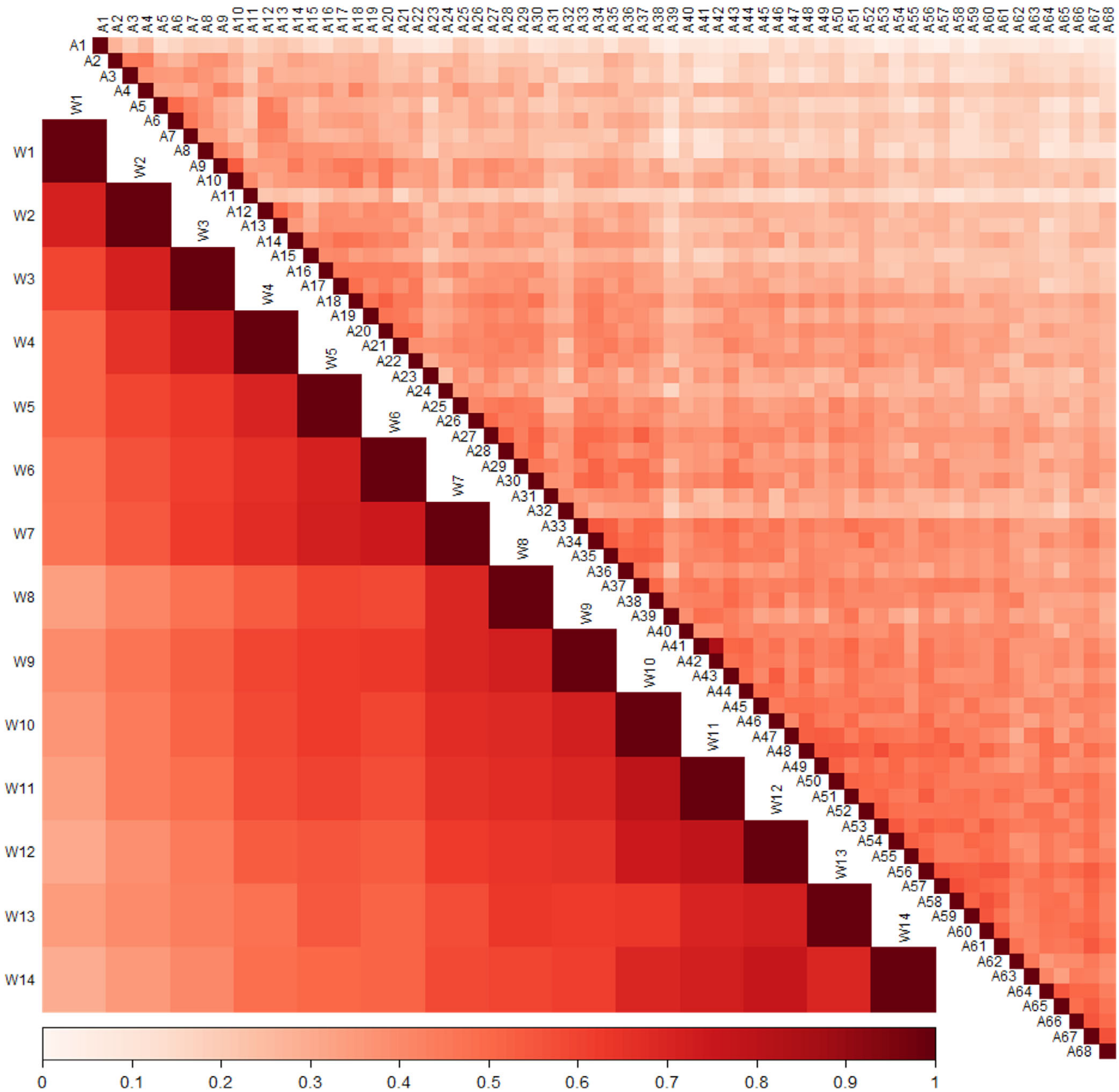

FIGURE 5 The Linear correlations among individual responses (upper right) and week-level aggregate measurements (lower left)

100 times and reported the average estimates. The estimates based on the two methods were close to each other. As expected, these estimates were higher than those based on the equal interval and equal length methods. The ω estimates based on the equal interval were quite low, suggesting substantial condition-specific effects. The estimates of this method were lower than those of equal length. The reason was that a student usually made a different number of submissions on different assignments. Two bins derived from equal length might contain submissions on the same assignment (see Figure 2), but two bins derived from equal interval did not. Thus, the condition-specific effect was less severe in the equal length method than in the equal interval method.

The correlations among bin-level indicators of the same week (Figure 7) further illustrated the condition-specific effects. Generally, these correlations based on the equal interval method were lower than those of equal length, which in turn were lower than random allocation and condition-representative methods.

5 | DISCUSSION

Aggregating process data is a common practice in LA. This paper explores the pros and cons of this practice in terms of psychometrics. We argue that aggregate measurements over a relatively short period

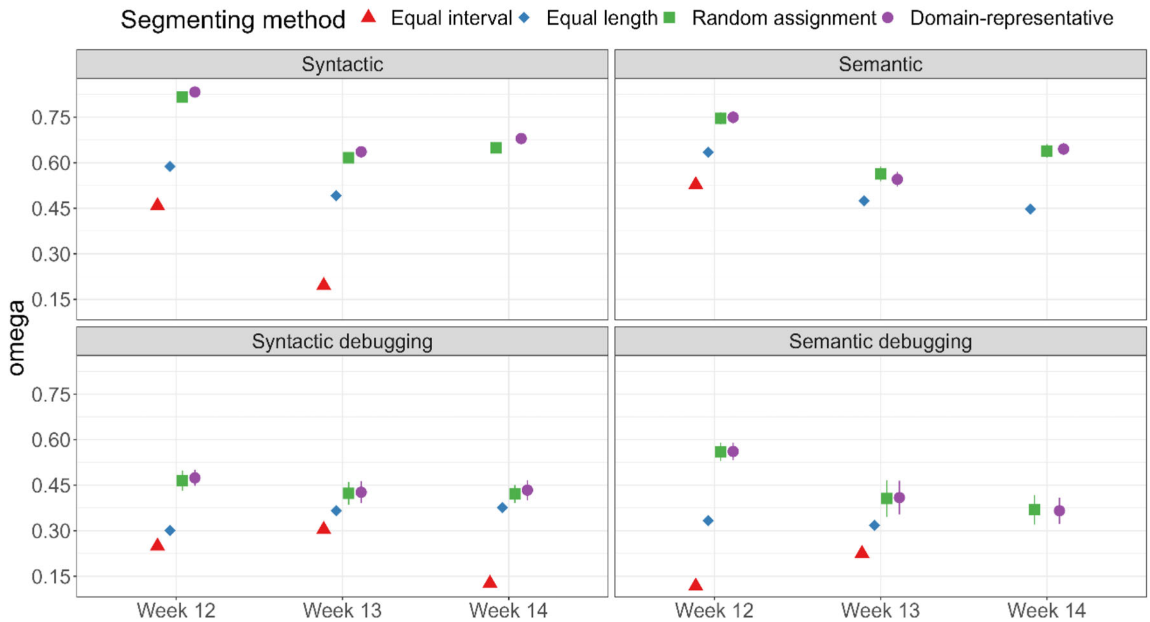

FIGURE 6 Omega of programming proficiency indicators computed by different segmenting methods. Error bars represent the ±1 standard deviations of ω estimates in the 100 repetitions for random allocation and condition-representative. The estimates were missing for equal interval and length methods in a few cases because the confirmatory factor model could not converge.

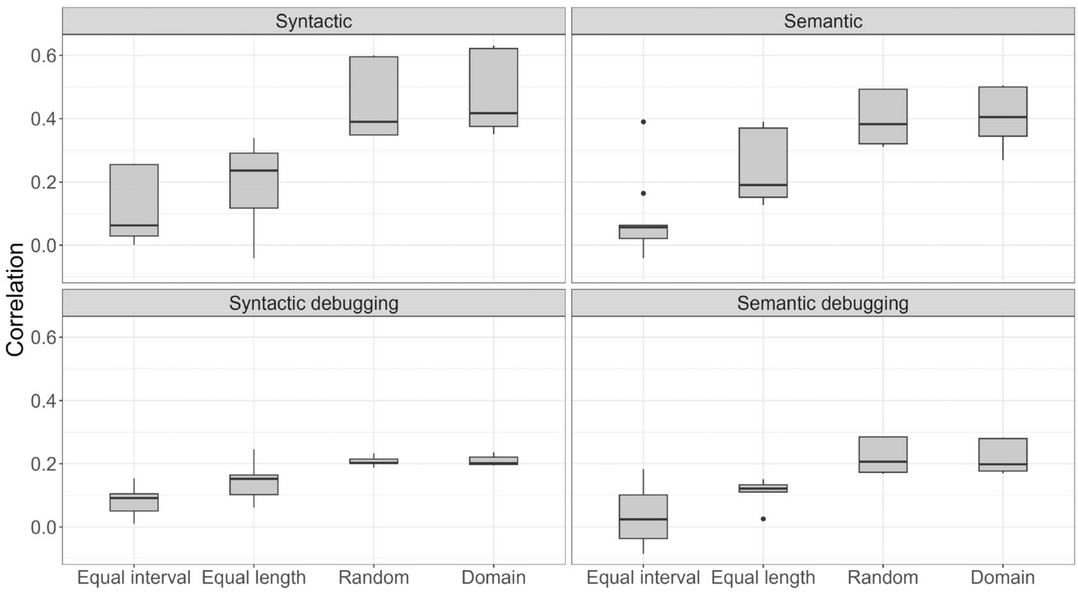

FIGURE 7 Linear correlations among bin-level programming proficiency indicators of the same week. Random: random allocation. Condition: condition-representative.

are more reliable than individual events, but aggregation over a relatively long period may mask meaningful changes in learning behaviours and should be avoided. The main contributions of this study are twofold: (1) we presented a general guideline for computing reliability metrics; (2) we conducted a case study to demonstrate the guideline. More broadly, this paper may foster the development of standards for rigour in LA research (Knight et al., 2019), mainly statistical and psychometric rigour.

5.1 | Interpreting reliability coefficients

Reliability coefficients quantify the proportion of random errors in the observed measurement. There is no universal threshold that we can use to assign a qualitative label to measurements, such as “poor” or “adequate”. Taking ICC as an example, a widely adopted guideline is that ICC values smaller than 0.5, between 0.5 and 0.75, between 0.75 and 0.9, and greater than 0.9 indicate poor, moderate, good, and excellent reliability (Koo & Li, 2016). However, as the authors of this guideline warn, "there are no standard values for acceptable reliability using ICC" (Koo & Li, 2016, p. 158). For instance, a recent study found that the median ICC of 37 behavioural tasks for measuring self-regulation was only 0.311 (Enkavi et al., 2019). Based on the guideline, over half of these behavioural tasks would be regarded as poor in test–retest reliability. However, the reason for low ICC may be that behavioural tasks are often selected for generating scores with low between-subject variance (Hedge et al., 2018) rather than that these behavioural tasks are truly unreliable. Therefore, compared with interpreting a reliability estimate based on a general threshold or guideline, the interpretation would be more meaningful when it is situated in the domain-specific spectrum of the estimates (Parsons et al., 2019), which should cover various samples and learning environments. Nevertheless, establishing the spectrum relies on a LA research culture where reporting reliability estimates for aggregate measurements of process data is a standard practice.

5.2 | Handling aggregate measurements with low reliability

Assuming that the guideline is applicable, when an aggregate measurement has a low reliability estimate, say 0.5, researchers may lengthen the period aggregated to increase the reliability, as suggested by Figure 3. However, we want to highlight the "embracing imperfection" notion in LA (Kitto et al., 2018). LA is an interdisciplinary community. The implicit assumption of psychometrics that improving measurement reliability is a worthy effort may not always hold in LA. In cases where increasing the period aggregated would mask behavioural adaptation in learning, it may be acceptable to embrace an imperfect measurement.

To maintain the psychometric rigour, instead of lengthening the period aggregated, we recommend reinspecting the assumption that the construct of interest is stable within the period. Researchers may examine this assumption by conducting a paired-sample *t*-test or repeated analysis of variance (ANOVA) to compare the measurements of different bins. If the assumption does not hold, decomposing the period may be considered. If the assumption holds, it means that even though the construct of interest is stable, there are still non-negligible measurement errors. Then, latent variable modelling can be used to account for the measurement errors in the analyses of the aggregate measurements and their relationships with other variables (Skrondal & Rabe-Hesketh, 2004).

5.3 | Practical implications

Beyond the potential of improving LA research practice, this study has implications for designing learning dashboards, specifically for designing action-related indicators (Schwendimann et al., 2017). These indicators usually show users aggregate information about learners' actions for self-monitoring, monitoring peers, or administrative monitoring. No matter for which purpose, the period being aggregated should not be too short; otherwise, the indicator would be unreliable and contain so much noise that regulation effort and feedback based on them would be inappropriate. Meanwhile, the period should not be too long to hide temporal patterns in learners' activities. In addition, indicators aggregated over a long period may not accurately reflect learners' current cognitive, motivational, and affective states. Instructional interventions based on such indicators may be ineffective due to low adaptivity (Aleven et al., 2017).

5.4 | Limitations and future directions

The current study focuses on the reliability of measurements derived from learning process data. We do not advocate simply pursuing high reliability because other measurement issues, such as validity and generalization, hold the same importance. A measurement with high reliability in a particular context does not imply that it is a valid indicator of the construct of interest and that it generalizes to another context. Indeed, Du et al. (2023) found that generalization and validity are two key challenges in using process data to measure self-regulated learning. Further research is suggested to provide reliability evidence as well as validity and generalization evidence for aggregate measurements of process data.

Some aggregate measurements that characterize the co-occurrence and sequential characteristics of events are association strength metrics, such as transition metrics, which quantify the tendency of one event following another (Bosch & Paquette, 2021). Regression models can estimate and account for their measurement errors (Matayoshi & Karumbaiah, 2021). Thus, it may be unnecessary to apply the proposed method for computing the reliability coefficients of these measurements.

Deciding the period length or granularity for aggregation relies on the research questions and contexts. Identifying the best length will maximize the reliability of measurements and minimize the risk of masking changes in learning behaviours. However, it may be challenging to identify the best length. There are methods for determining the minimum observations for deriving an aggregate measurement that is stable and representative of the population (Hehman et al., 2018). Such methods cannot directly apply to process data in LA because the aggregation is over subjects rather than responses from the same subjects. Alternatively, researchers may plot the reliability metrics against the length of the aggregated period, like in Figure 3, and identify the length point after which the increasing rate of reliability metrics slows down. However, it may be challenging to identify such a point, and the period length at this point may be large (e.g., 20 responses in Figure 3). Further research is needed to investigate the utility of this method and develop other techniques for determining the best length for aggregation.

Random allocation and condition-representative segmenting demand repetitions to reduce biases caused by randomness. The 100 repetitions for computing the reliability estimates of programming proficiency took one hour. A guideline for the minimum repetitions avoids unnecessary repetitions and saves time. Yet, various features of the data, such as the process length, the reliability of individual events, and the variation in the learning conditions, may influence the minimum repetitions. It entails extensive simulation experiments and empirical data analyses to yield the guideline.

AUTHOR CONTRIBUTIONS

Yingbin Zhang: Conceptualization; methodology; formal analysis; writing – review and editing; writing – original draft; visualization; investigation; project administration; resources; supervision. **Yafei Ye:** Conceptualization; methodology; writing – review and editing; writing – original draft; investigation. **Luc Paquette:** Conceptualization; methodology; writing – review and editing; supervision; funding acquisition; project administration; resources. **Yibo Wang:** Methodology; writing – review and editing; formal analysis. **Xiaoyong Hu:** Conceptualization; writing – review and editing; resources.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The use of the data in this study for research purposes was approved by the Institutional Review Boards (IRB) of the University of Illinois at Urbana-Champaign.

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